

Lecture 6: Regression, Hypothesis Testing & Optimization

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People often think AI, machine learning, complicated models are the most common tool for quants. Yet, it is not the case in the industry. Basic statistical methods and testing are still predominant methods utilized in the industry to discover alpha. This lecture will cover basics of the practical tools being used in work from day-to-day.

Please do not hesitate to reach me via WhatsApp, either in the group or PM, if you have any questions regarding the course.

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1 Linear Regression

1.1 Introduction

Given a simplest example, we would like to build a model to estimate one's salary by their age:

$$\text{Salary} = m + \text{age} \times \beta \quad (1)$$

In this example, there are three components in this equation:

- Salary: Response
- c: Intercept
- age: inputs
- β : Slope

It is also possible to extend the model to support multiples inputs:

$$\text{Salary} = c + \text{age} \times \beta_1 + \text{education} \times \beta_2 + \frac{1}{\text{University Ranking}} \times \beta_3 \quad (2)$$

Regardless of the form of the inputs, linear regression can be always written in the form of

$$y = \beta_0 + \sum_i x_i \beta_i \rightarrow \mathbf{Y} = \vec{\beta} \mathbf{X} \quad (3)$$

One have been studying machine learning will know that, there are different models cater for different prediction tasks, namely regression and classification. For instance, if we would like to predict whether a person is employed or not using the same inputs right after graduation, we may build a logistic regression:

$$\log \frac{p}{1-p} = \beta_0 + \beta_{age} \cdot \text{age} + \beta_{education} \cdot \text{education} + \beta_{ranking} \cdot \frac{1}{\text{ranking}} \quad (4)$$

After rearranging the formula,

$$P(\text{employed}|X) = \frac{e^{\beta x}}{1 + e^{\beta x}} \quad \text{where } e^{\beta x} \in [0, \infty] \quad (5)$$

where p is the probability that one is employed or not.

1.2 Estimation Method

Given a set of observed output (e.g. salary) $\mathbf{y}$ and observed inputs (e.g. ages, education, rankings) $\mathbf{x}$, we would like to find β s.t. it minimize the error between the model and the actual observations.

$$\vec{\hat{\beta}} = \underset{\vec{\beta}}{\operatorname{argmin}} L(D, \vec{\beta}) = \underset{\vec{\beta}}{\operatorname{argmin}} \sum_{i=1}^n (\vec{\beta} \cdot \vec{x}_i - y_i)^2 \quad (6)$$

The loss function $L(D, \vec{\beta})$ can be written as

$$L(D, \vec{\beta}) = \|\mathbf{X}\vec{\beta} - \mathbf{Y}\|^2 = \mathbf{Y}^T\mathbf{Y} - \mathbf{Y}^T\mathbf{X}\vec{\beta} - \vec{\beta}^T\mathbf{X}^T\mathbf{Y} + \vec{\beta}^T\mathbf{X}^T\mathbf{X}\vec{\beta} \quad (7)$$

By taking the gradient of the loss function to zero,

$$\frac{\partial L(D, \vec{\beta})}{\partial \vec{\beta}} = -2\mathbf{X}^T\mathbf{Y} + 2\mathbf{X}^T\mathbf{X}\vec{\beta} = 0 \rightarrow \vec{\hat{\beta}} = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{Y} \quad (8)$$

2 Generalized Linear Model

2.1 Motivation

Above example illustrates that, even for classification problem, it can be easily formulate as a linear regression model after applying some transformation.

Let's take a look at the regression problem (1) above. Regardless of the variables, result is a continuous value. To be more precise, the regression formula should be written as:

$$\text{Salary} = m + \text{age} \times \beta_1 + \text{education} \times \beta_2 + \frac{1}{\text{University Ranking}} \times \beta_3 + \epsilon \quad (9)$$

where the ϵ is a random variable that usually assumed to follow normal distribution. Therefore, for a general regression problem

$$\mathbf{Y} = \vec{\beta}\mathbf{X} + \vec{\epsilon} \quad (10)$$

The model does not do prediction with certainty; the model itself has assumed errors in it. Yet, when we are doing observation or prediction, we mostly care about the "mean" of the prediction result. Assuming the prediction model is not biased (i.e. $E(\epsilon) = 0$)

$$E(\mathbf{Y}|\mathbf{X}) = \vec{\beta}\mathbf{X} \quad \text{where } \mathbf{Y} \sim \text{Normal Distribution} \quad (11)$$

Now, let's look at the classification problem in (5).

$$E(\mathbf{Y}|\mathbf{X}) = \vec{\beta}\mathbf{X} \quad \text{where } \mathbf{Y} \sim \text{Binomial Distribution} \quad (12)$$

2.2 Formulation

To generalize the above observation, we may see that for different modeling problem, by assuming a different distribution on $\mathbf{Y}$, the linear regression can be used to solve different type of problems.

Let y be the target or observation we are interested in (e.g. salary, employment probability), it can be seen that there exist a function $g(y)$ s.t.

$$g(y) = \beta_0 + \sum_i \beta_i x_i \quad (13)$$

can be used for modeling different types of problems.

For instance, when we are trying to predict a continuous variable like salary (i.e. $Y|X$ follow normal distribution), we may set $g(y) = y$ s.t. $y = \beta_0 + \sum_i \beta_i x_i$ (typical regression model).

When we are trying to predict a probability (like employment) (i.e. $Y|X$ follow a binomial distribution), we may set $g(y) = \log \frac{y}{1-y}$.

Here is the formal structure of generalized linear models:

1. If $y|\mathbf{x}$ follow an exponential family distribution $f(y; \theta; \phi)$ (e.g. normal distribution, binomial distribution)
2. There would exist a function $g(\mu)$ which is called the link function s.t.
3. For a linear predictor: $\eta = \beta_0 + \sum_i \beta_i x_i = \mathbf{x}'\beta$
4. The link function $g(\mu)$ links between the mean of response $\mu = E(y|\mathbf{x})$ and the linear predictor: $g(\mu) = \mathbf{x}'\beta$

You see there is no any assumption on the form and distribution of $\mathbf{x}$.

Distribution	P_θ	φ	θ	$b(\theta)$	$\mu = b'(\theta)$	$V(\theta) = b''(\theta)$
Binomial	$b(n_0, \pi)$	1	$\log \left(\frac{\pi}{1-\pi} \right)$	$n_0 \log(1 + e^\theta)$	$\frac{n_0}{(1 + e^{-\theta})}$	$\mu \left(1 - \frac{\mu}{n_0} \right)$
Neg-Binomial	$nb(r_0, \pi)$	1	$\log(1 - \pi)$	$-r_0 \log(1 - e^\theta)$	$\frac{r_0}{e^{-\theta} - 1}$	$\mu \left(1 + \frac{\mu}{r_0} \right)$
Poisson	$\wp(\lambda)$	1	$\log(\lambda)$	e^θ	e^θ	μ
Gaussian	$N(\mu, \sigma^2)$	σ^2	μ	$\theta^2/2$	θ	1
Gamma	$\Gamma(r, \lambda)$	$1/r$	$-1/\mu$	$-\log(-\theta)$	$-1/\theta$	μ^2
Inv-Gaussian	$IG(\mu, \lambda)$	$2/\lambda$	$-1/\mu^2$	$-2\sqrt{-\theta}$	$1/\sqrt{-\theta}$	$\mu^3/2$

Figure 1: For different y distribution, here are their corresponding link function θ .

3 Hypothesis Testing

3.1 Motivation

Why we need hypothesis testing? Imagine you have a coin. You want to test if the coin is fair. In order to quantify the condition to "accept" or "reject" the coin is fair, you need to define a boundary condition of when you should make the call.

Another scenario that you would need to do hypothesis testing is comparing the performance of two strategies. You collect the performance of two strategies across multiple trading periods. You would like to test if the mean return of the first strategy μ_1 is larger or smaller than the return of the second strategy μ_2 .

Let's focus on the coin flipping example. Assume you did 3 coin flips, there are 8 different possible outcomes: $\{HHH\}, \{HHT\}, \{HTH\}, \{HTT\}, \{THH\}, \{THT\}, \{TTH\}, \{TTT\}$. The question is, which of the outcome above we should consider the coin is "fair", and which of the outcome should be considered as "unfair".

Formally speaking, we would like to test the hypothesis that is $\theta = \frac{1}{2}$ where θ is the probability of head. The hypothesis that one is interested in (null hypothesis: H_0) is often defined as Θ_0 and the opposite (alternative hypothesis: H_1) is defined as Θ_1 . For coin flipping, $\Theta_0 = \{\frac{1}{2}\}$ and $\Theta_1 = [0, \frac{1}{2}) \cup (\frac{1}{2}, 1]$.

$$\begin{cases} H_0 : & \theta = \frac{1}{2} \\ H_1 : & \theta \neq \frac{1}{2} \end{cases} \quad (14)$$

A hypothesis test is a function d that maps observations into a conclusion; either accept or reject.

$$R_{\mathbf{x}}^n \xrightarrow{d} \{a_0 : \text{accept } H_0, a_1 : \text{reject } H_0\} \quad (15)$$

$$d(x) = a_i \quad \text{for } x \in C_i \quad (16)$$

where C_0 is called the acceptance region of H_0 , and C_1 is called the rejection region of H_0 .

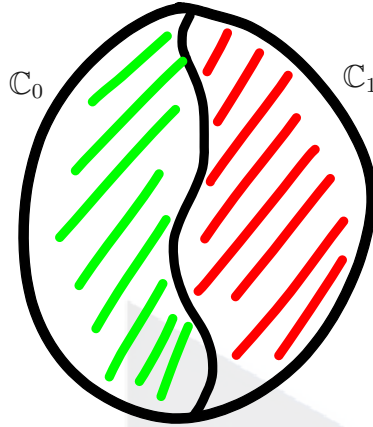


Figure 2: Sample space divided into acceptance and rejection region.

A question remains, what is the "best" d (i.e. C_0 and C_1)?

3.2 Error

To analyze the "performance" of a test, it is important to know what types of error are there.

	H_0 accepted ($x \in C_0$)	H_0 rejected ($x \in C_1$)
$H_0 (\theta \in \Theta_0)$ true	no error	type I error
$H_0 (\theta \in \Theta_1)$ false	type II error	no error

In a perfect world, we would like to have no type I and II error. However, for a statistical testing model, often one needs a trade off between type I and II error.

Conventionally, one would assign an upper bound to the type I error and to minimize the type II error given this constraint.

$$\underset{C_0, C_1}{\operatorname{argmin}} P(\mathbf{X} \in C_0) \quad \forall \theta \in \Theta_1 \quad (17)$$

subject to the condition that

$$\sup\{P(\mathbf{X} \in C_1 : \theta \in \Theta_0)\} = \alpha \quad (18)$$

where α is the significant level; $\alpha \in (0, 1)$

3.3 Power

Power function of a test is defined as

$$Q(\theta) = \int_{C_1} f_{\mathbf{X}}(x, \theta) \quad (19)$$

For $\forall \theta \in \Theta_1$, $Q(\theta)$ is called the power of the test. With the definition of power, we may rewrite (18) as,

$$\sup\{Q(\theta) : \theta \in \Theta_0\} = \alpha \quad (20)$$

To compare two tests with the same significant level α with power functions, we would say a test is more powerful than the second at a point θ^* of Θ_1 if $Q_1(\theta^*) > Q_2(\theta^*)$. A test is called uniformly most powerful if it is the most powerful at each $\theta \in \Theta_1$.

3.4 Uniformly Most Powerful Test

What is the intuitive understanding of power? It is a measure of how good a test in rejecting the hypothesis when the hypothesis is indeed wrong (i.e. $\theta \in \Theta_1$). Consider the following hypothesis:

$$\begin{cases} H_0 : \theta = \theta_0 & \Theta_0 = \{\theta_0\} \\ H_1 : \theta > \theta_0 & \Theta_1 = \{\theta : \theta > \theta_0\} \end{cases} \quad \text{where } \Theta = \{\theta : \theta \geq \theta_0\} \quad (21)$$

For the uniformly most powerful test at significant level α , it must satisfies the following condition:

1. $Q(\theta_0) = \alpha$
2. $Q(\theta) \geq Q_*(\theta) \quad \forall \theta \in \Theta_1$ where $Q_*(\theta)$ is any other test with significant level α

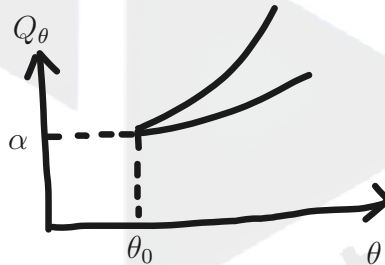


Figure 3: Sample space divided into acceptance and rejection region.

4 Linear Programming

4.1 Motivation

Linear programming is a very common problem in optimization. It often involves complicated mathematics for solving simple problems. With the development of programming library, it is not that important to have low level understanding of implementation of the algorithm. But, it is still important to understand the formulation of linear programming problems since libraries still requiring proficiency in that in order to use them.

Consider the following problem:

$$\begin{aligned}
&\text{minimize} && z = -x_1 - 2x_2 \\
&\text{subject to} && -2x_1 + x_2 \leq 2 \\
&&& -x_1 + x_2 \leq 3 \\
&&& x_1 \leq 3 \\
&&& x_1, x_2 \geq 0
\end{aligned}$$

For the problem, it can be divided into two parts: 1) Objective (minimize $z = -x_1 - 2x_2$), and 2) Constraints ($-2x_1 + x_2 \leq 2, -x_1 + x_2 \leq 3, \dots$).

4.2 Formulation

All linear optimization problem can be converted into standard form like the following:

$$\begin{aligned}
&\text{minimize} && z = c^T x \\
&\text{subject to} && Ax = b \\
&&& x \geq 0 \\
&\text{with} && b \geq 0
\end{aligned}$$

For instance,

$$\begin{aligned}
&\text{minimize} && z = 4x_1 - 5x_2 + 3x_3 \\
&\text{subject to} && 3x_1 - 2x_2 + 7x_3 = 7 \\
&&& 8x_1 + 6x_2 + 6x_3 = 5 \\
&&& x_1, x_2, x_3 \geq 0
\end{aligned}$$

is in standard form with the following matrix-vector form:

$$x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}, \quad c = \begin{bmatrix} 4 \\ -5 \\ 3 \end{bmatrix}, \quad A = \begin{bmatrix} 3 & -2 & 7 \\ 8 & 6 & 6 \end{bmatrix}, \quad b = \begin{bmatrix} 7 \\ 5 \end{bmatrix} \quad (22)$$

The standard form is a commonly required form of inputs for different linear optimization libraries.

4.3 Standard Form Conversion

Often linear optimization problems is not directly in standard form. Different tricks can be applied to convert problems into standard form as a programming library input.

Maximization Problem

For a maximization problem, it can easily be converted into a minimization problem by multiplying the objective by -1. For example,

$$\text{maximize } z = 4x_1 - 3x_2 + 6x_3 = c^T x \rightarrow \text{minimize } \hat{z} = -4x_1 + 3x_2 - 6x_3 = -c^T x \quad (23)$$

Constraint Sign

If any of the component of b is negative, those constraints can be multiplied by -1 to change the sign. For example,

$$x_1 - 2x_2 \leq -2 \rightarrow -x_1 + 2x_2 \geq 2 \quad (24)$$

Non-zero lower bound

If a variable has a lower bound that is not zero, we may replace the variable in order to achieve a zero lower bound.

$$x_1 \geq 5 \rightarrow x'_1 = x_1 - 5 \rightarrow x'_1 \geq 0 \quad (25)$$

Unrestricted variable

For variables without specified lower or upper bounds, they are called *free or unrestricted variable*. They can be replaced by a pair of non-negative variables. For example, if x_2 is a free variable,

$$x_2 = x'_2 - x''_2 \quad \text{with} \quad x'_2, x''_2 \geq 0 \quad (26)$$